```
> restart;#Sun
```

```
> g := 3 R^2 / (14 π) r^(-2) - 1/4 R/π r^(-1) + 1/(14 π) ;
```

$$g := \frac{3\,R^2}{14\,\pi\,r^2} - \frac{R}{4\,\pi\,r} + \frac{1}{14\,\pi} \tag{1}$$

```
> R := 4.235518529·10^22; NAv := 6.02214076·10^26; Mp := 5.9722·10^24; Mnut :=
  1.674027498 × 10^-27; Mprot := 1.67262192359·10^-27; Nneut := Mp/Mnut;
  Nprot := Mp/Mprot; Newton := NAv·0.102; 0.102/Mnut; 0.102/Mprot; Rp := 6371000;
  G := 6.6743·10^-11; Newton1 := 1.161068739 × 10^25; NrNucEarth := NAv
  ·19.28·Mp; Msun := 1.9885 × 10^30;   Nnuc := 19.28· 8.38 × 10^49;
  Nsun := 1.198 × 10^57; Rsun := 6.9634 × 10^8; MassEarthSunC :=
  Nnuc·Nsun·G·(1.672621923·10^-27)^2/(15·10^10); FclasEthC := 1.007622549 × 10^-77/Rp^2;
  FclasEth := G·(1.672621923·10^-27)^2/Rp^2; Nnuc·Newton/Rp^2·G; Mmoon := 7.35
  ·10^22; Nmoon := Mmoon/Mprot; NmoonT := (0.42 × 16) + (0.21 × 28) + (0.07 × 24)
  + (0.08 × 56) + (0.06 × 40) + (0.04 × 27) + 0.12·24; Nmoon1 := NmoonT/100
  ·7.35·10^22·NAv;
  #Number of nucleons in 102 g and nucleons into Earth;&
  Clasical Gravitational Force between 2 nucleons approx. Nr. of
  nucleons in 102 g and Earth at radius rp.  # NAv = Avogadro
  number 6.02214076·10^26
  # Considering the Earth's crust is composed of approximately
  47 % oxygen and 28 % silicon (by mass),
  #we can estimate the average number of nucleons per atom as
  follows:Average nucleons per atom = (0.47·16 + 0.28 ·28) = 15.56
  #but approx ~19.28 nuleons per atom counting hevy atoms and
   8.38 × 10^49 atoms inside Earth ;
    Nsun is the number of proton
  # in the Sun # The Moon's crust composition is: Oxygen (O):
  42%  Silicon (Si) : 21 % Magnesium·(Mg) : 7 % Iron·(Fe) : 8 %
  Calcium·(Ca) : 6 % Aluminum (Al) : 4 % & 12 % other elements =
  > ~7.35 × 10^22 kilograms• Averige Mass·NAv
```

$$R := 4.235518529 \times 10^{22}$$

$$NAv := 6.022140760 \times 10^{26}$$

$$Mp := 5.972200000 \times 10^{24}$$

$$Mnut := 1.674027498 \times 10^{-27}$$

$$Mprot := 1.672621924 \times 10^{-27}$$

$$Nneut := 3.567563858 \times 10^{51}$$

$$Nprot := 3.570561831 \times 10^{51}$$

$$Newton := 6.142583575 \times 10^{25}$$

$$6.093089876 \times 10^{25}$$

$$6.098210153 \times 10^{25}$$

$$Rp := 6371000$$

$$G := 6.674300000 \times 10^{-11}$$

$$Newton1 := 1.161068739 \times 10^{25}$$

$$NrNucEarth := 6.934134721 \times 10^{52}$$

$$Msun := 1.988500000 \times 10^{30}$$

$$Nnuc := 1.615664000 \times 10^{51}$$

$$Nsun := 1.198000000 \times 10^{57}$$

$$Rsun := 6.963400000 \times 10^{8}$$

$$MassEarthSunC := 2.409449900 \times 10^{33}$$

$$FclasEthC := 2.482462333 \times 10^{-91}$$

$$FclasEth := 4.600299244 \times 10^{-78}$$

$$1.631896593 \times 10^{53}$$

$$Mmoon := 7.350000000 \times 10^{22}$$

$$Nmoon := 4.394298493 \times 10^{49}$$

$$NmoonT := 25.12$$

$$Nmoon1 := 1.111879893 \times 10^{49} \quad \textbf{(2)}$$

>

```
#Number of nucleons in 102 g and nucleons into Earth;&
Clasical Gravitational Force between 2 nucleons approx. Nr. of
nucleons in 102 g and Earth at radius rp.  # NAv = Avogadro
number 6.02214076·10^26
# Considering the Earth's crust is composed of approximately
```

```
    47 % oxygen and 28 % silicon (by mass),
    #we can estimate the average number of nucleons per atom as
    follows:Average nucleons per atom = (0.47·16 + 0.28 ·28) = 15.56
    #but approx ~19.28 nuleons per atom counting hevy atoms and
     8.38 × 10^49 atoms inside Earth ;
```

```
> Mp; 3.567563858 × 10^51·Mnut; Nnuc·Nsun·G·(Mnut)^2/(15·10^10)^2; Nprot· Nsun·G·(Mnut)^2/(15·10^10)^2;
  Mmoon; Nmoon1·Nnuc·G·(Mnut)^2/(384400 ·10^3)^2; Nmoon·Nnuc·G·(Mnut)^2/(384400 ·10^3)^2;
  # 15·10^10 the distance Sun − Earth, we evaluate the 384400
  ·10^3 the distance between Earth − Moon
```

$$5.972200000 \times 10^{24}$$

$$5.972199999 \times 10^{24}$$

$$1.609000751 \times 10^{22}$$

$$3.555836282 \times 10^{22}$$

$$7.350000000 \times 10^{22}$$

$$2.273908922 \times 10^{19}$$

$$8.986793093 \times 10^{19} \quad (3)$$

```
> Nnuc*Newton; (Nnuc*Newton)^(-1);
  Nnuc*Newton*subs(r = Rp, 4.089903752*10^(-64)/r^2);
```

$$9.924351149 \times 10^{76}$$

$$1.007622549 \times 10^{-77}$$

$$1.000000000 \quad (4)$$

```
> # And fitting the data we have:
```

```
> V := 10^14/2.463682791 · (1.007622549 × 10^-77/r^2 − 3.295653603 × 10^-100/r
    + 2.639796229 × 10^-123) ;
```

$$V := \frac{4.089903752 \times 10^{-64}}{r^2} - \frac{1.337693966 \times 10^{-86}}{r} + 1.071483812 \times 10^{-109} \quad (5)$$

```
> V1 := 10^14/(2.209965582·2.463682791) · (1.007622549 × 10^-77/r^2
```

$$-\frac{3.295653603\times10^{-100}}{r}+2.639796229\times10^{-123}\Bigg);$$

$$V1 := \frac{1.850664004\times10^{-64}}{r^2}-\frac{6.053008141\times10^{-87}}{r}+4.848418551\times10^{-110} \quad (6)$$

```
> Newton·Nnuc·subs(r = Rp, 4.089903752×10^-64/r^2); Gf := Newton·Nprot
  ·subs(r = Rp, V); Gf1 := Newton·Nprot·subs(r = Rp, V1); Newton·Nsun
  ·subs(r = Rsun, V1);
  # 1kg on the Earth weight 9.81 Newtons coresponding to Nnuc·1
  Newton  averrage nucleons
```

$$1.000000000$$

$$Gf := 2.209965582$$

$$Gf1 := 0.9999999999$$

$$28.08618704 \quad (7)$$

```
> # The force between 2 nucleons at radius Earth from atoms in the
  Sun and Earth
```

```
> Gsun_ear := Newton·Nsun·V1;
```

$$Gsun_ear := \frac{1.361869426\times10^{19}}{r^2}-\frac{0.0004454296785}{r}+3.567861576\times10^{-27} \quad (8)$$

```
> 1/(NrNucEarth·Newton); 1/(Nnuc·Newton);
```

$$2.347775957\times10^{-79}$$

$$1.007622549\times10^{-77} \quad (9)$$

```
> # This is the analitical calculus below :
> # The force between 2 nucleons at radius Earth from atoms on
  ground level analitical calculus
```

```
> h := evalf(3 R^2/(14 π r^2) - R/(4 π r) + 1/(14 π));
```

$$h := \frac{1.223648038\times10^{44}}{r^2}-\frac{3.370518552\times10^{21}}{r}+0.02273642044 \quad (10)$$

```
>
> Gfn := Newton·Nprot·V1/(9.924351149×10^76); Newton·Nprot·subs(r = Rp, V1);
  # The attraction forces between ~ 10^77 nucleons at distance Rp
  is 1 Newton
```

$$Gfn := \frac{4.089903753 \times 10^{-64}}{r^2} - \frac{1.337693966 \times 10^{-86}}{r} + 1.071483812 \times 10^{-109}$$

$$0.9999999999 \quad (11)$$

```
> Nnuc·Newton; (Nnuc·Newton)^-1; Nnuc·Newton ·subs(r = Rp,
    4.089903752 × 10^-64 / r^2);
    # the clasical constant is ~1.161043022×10^-121
```

$$9.924351149 \times 10^{76}$$

$$1.007622549 \times 10^{-77}$$

$$1.000000000 \quad (12)$$

```
> subs(r = 4.2·10^-15, Gfn);
    # Force in Newton between 2 nucleons at 4.2 fm
```

$$2.318539542 \times 10^{-35} \quad (13)$$

```
> subs(r = 2.4·10^-15, Gfn);
    # Force in Newton between 2 nucleons at 2.4 fm
```

$$7.100527349 \times 10^{-35} \quad (14)$$

```
>  subs(r = 6371000, Gfn);
    # Force in Newton between 2 nucleons at radius Earth
```

$$1.007622549 \times 10^{-77} \quad (15)$$

```
> #1.87 × 10^-64 / r^2 < ÷ > 4.089903752 × 10^-64 / r^2;
    # A relatively small distance less than 10^15"m" we approximate
    the attraction between 2 nucleons  clasic calculus;
> k := 6.67408·10^-11 ;#Gravitational constant in N
```

$$k := 6.674080000 \times 10^{-11} \quad (16)$$

```
> subs(r = 4.2·10^-15, k·Mnut·Mnut / r^2);
```

$$1.060273732 \times 10^{-35} \quad (17)$$

```
> subs(r = 2.4·10^-15, k·Mnut·Mnut / r^2);
```

$$3.247088305 \times 10^{-35} \quad (18)$$

```
>  2.318539542 × 10^-35 / (1.060273732 × 10^-35);
```

$$2.186736757 \quad (19)$$

```
> 4.089903752 × 10^-64 / 2.186736757;
```

$$1.870322863 \times 10^{-64} \tag{20}$$

```
>
```